Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/07/2021
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	05/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 1/8/19) NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The initial working uncertainty budget projected an accuracy of 2.2×10^(-16). The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy. The PI will also participate in a study on space optical clocks as a member of NASA's science definition team (SDT) for optical clocks and time transfer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British, Canadian, South Korean, Polish, and Indian National Laboratories. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is being used as a correction of International Atomic Time, including the correction for the PHARAO clock. More broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks, including testing fundamental physics with state-of-the-art optical clocks in space.
Task Progress:	With the PHARAO instrument team, we published a pre-launch accuracy evaluation of the microgravity PHARAO clock. The paper significantly improves PHARAO's expected accuracy on-orbit and shows that it can meet its accuracy goal after 18 months of operation, and below 1 part in 10^16 for comparisons with cesium ground clocks. The evaluation procedure developed in the paper is a new approach, accurately evaluating the correction for PHARAO operating at a nominal density and launch velocity by operating at several optimized launch velocities and cold atom densities. This nominal operating point provides PHARAO a high frequency stability for dedicated approximately 10-day long comparisons with ground clocks. The high accuracy and stability of PHARAO is needed for the mission objective of ACES, to test general relativity, to probe the time variation of fundamental constants, and to compare high-performance ground clocks worldwide. The Penn State group also developed a procedure to characterize the profile of the microwave field in the state preparation cavity during the on-orbit commissioning phase of PHARAO. The variations of the amplitude in the state preparation cavity changes the density distribution of the cold atoms and thereby can impact a number of systematic frequency errors of the PHARAO clock, including Doppler shifts, the cold collision frequency shift, and the microwave lensing frequency shift. The Principal Investigator (PI) Gibble is also a member of the NASA science definition team (SDT) for optical clocks and time transfer. The team prepared a Science Envelope Requirements Document (SERD) for optical clocks in space.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Peer-reviewed Journals	Laurent Ph, Esnaut FX, Gibble K, Peterman P, Lévèque T, Delaroche Ch, Grosjean O, Moric I, Abgrall M, Massonnet D, Salomon Ch. "Qualification and frequency accuracy of the space based primary frequency standard PHARAO." Metrologia. 2020 Oct;57(5):055005. https://doi.org/10.1088/1681-7575/ab948b , Oct-2020
NASA Technical Documents	Oates C, Gibble K, Hollberg L, Newbury N, Yu N. "Science Envelope Requirements Document (SERD) for A Vision for a State-of-the-Art Optical Clock in Space." NASA technical publication, September 2020. , Sep-2020